



FlockOS-based Integrations Empower Organisations to Communicate Flawlessly

FlockOS, which is touted to be the world's first chat operating system, is largely based on open source technologies. Starting from HTML, CSS and JavaScript, to JVM, MongoDB and MySQL, the platform has a vast number of community solutions to enable real-time communication.

"Open source technologies have played an instrumental role in the evolution of FlockOS," says Bhavin Turakhia, CEO and founder of Flock. Thirty-seven-year-old Turakhia is renowned for his successful startup innovations in the Indian business landscape with ventures such as Radix, CodeChef Zeta, Ringo and team messenger platform Flock, chose open source to make FlockOS a standout offering.

In order to ensure developers build customised apps, bots and integrations on Flock, the product team designed the concept of FlockOS. FlockOS helps developers build workspace apps that can be used within Flock, the messaging and collaboration tool.

Apps as first-class citizens

While Flock is supported across all leading platforms—including Windows, Mac, iOS, Android and the Web, its FlockOS requires a standalone platform to allow apps to blend seamlessly with the native Flock experience.

"We wanted to build a platform that

is extremely powerful, and where apps are treated as first-class citizens. To achieve both the goals, we needed to allow app developers to extend the capabilities of Flock in such a way, that they could show their own customised UI inside it," explains Chaitanya Gupta, chief technology officer, Flock.

Gupta and his small team of engineers initially searched for a framework that could work across platforms. Eventually, the development squad realised that the best fit for fast moving business teams were the Web technologies.

"The model we finally settled on was to use iframes on the desktop and the Web, and Web views on the mobile to present an app's user interface. We also had to work out how the Flock client cloud communicated with the app's interface and vice versa," Gupta tells *Open Source For You*.

The team has leveraged CSS, HTML and JavaScript to build the Windows and Mac presence of FlockOS, whereas its iOS presence was built with Objective-C and Swift. The Android variant, on the other hand, is entirely based on Java.

At the backend, Flock engineers deployed database solutions such as MongoDB, MySQL, Elasticsearch, Kibana and HBase. The company also used JVM to run Java code, while the Apache Spring framework smoothened the overall Web experience.

Aiding work at a faster pace

FlockOS uses multiple open source technologies and hence, opens infinite possibilities for scalability. The platform also enables users to access and accomplish all their work-related activities from a single place. "FlockOS allows users to save precious time, which might have otherwise been spent in shuffling from one office Intranet portal to another," says Ninad Raval, VP-product and design, at Flock.

In addition to letting developers build fresh apps on FlockOS, the platform also supports integration with other Web services. For instance, an Uber app on FlockOS lets a user book a cab while chatting with a dedicated team. Buttons can also be a part of the user panel, since they can be meticulously embedded in the interface and act as triggers, boosting app usage.

"Apps built using FlockOS can help users complete frequent tasks